Math 242, S09
Exam 2

Name: Answer
SSN: xxx-xx-

Instructions:

- There are totally 4 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	25	
2	25	
3	25	
4	25	
Total	100	

Good Luck !

Problem 1. (25 points) Given the following initial value problem (Logistic equation):

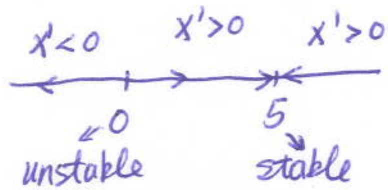
$$\frac{dx}{dt} = 15x - 3x^2, \qquad x(0) = 10$$

(a) Find all critical points of the given differential equation and then determine whether it is stable or unstable for each of them.

Solution: $15x - 3x^2 = 0$

$x(15-3x) = 0 \Rightarrow 3x(5-x) = 0$

$\Rightarrow x = 0,\ x = 5 \rightarrow$ critical points



(b) Solve the initial value problem for $x(t)$.

Solution: $\frac{dx}{dt} = 3x(5-x)$

$$\int \frac{1}{x(5-x)}\,dx = \int 3\,dt.$$

$$\frac{1}{5}\int\left(\frac{1}{x} + \frac{1}{5-x}\right)dx = \int 3\,dt.$$

$$\ln|x| - \ln|5-x| = 15t + C$$

$$\ln\left|\frac{x}{5-x}\right| = 15t + C.$$

$$\frac{x}{5-x} = e^{15t+C}$$

$$\Rightarrow \frac{x}{5-x} = Ae^{15t}$$

$$x(0) = 10 \Rightarrow A = \frac{10}{5-10} = -2$$

$$\Rightarrow \frac{x}{5-x} = -2e^{15t}$$

$$x = -10e^{15t} + 2e^{15t}x$$

$$\Rightarrow x(t) = \frac{10e^{15t}}{2e^{15t}-1} = \frac{10}{2-e^{-15t}}$$

Problem 2. (25 points) Given an initial value problem

$$\frac{dy}{dx} = y + x + 1, \qquad y(0) = -1.$$

Apply **Euler's method** to approximate the solution of the initial value problem on the interval $[0,\ 0.3]$ with step size $h = 0.1$.

Solution: $f(x,y) = y + x + 1, \quad x_0 = 0,\ y_0 = -1,\ h = 0.1.$

$x_1 = x_0 + h = \underline{0.1}$

$y_1 = y_0 + f(x_0, y_0)\cdot h = y_0 + (y_0 + x_0 + 1)\cdot h$

$= -1 + (-1 + 0 + 1)\cdot(0.1)$

$= \underline{-1}$

$x_2 = x_1 + h = 0.1 + 0.1 = 0.2.$

$y_2 = y_1 + f(x_1, y_1)\cdot h = y_1 + (y_1 + x_1 + 1)\cdot h$

$= -1 + (-1 + 0.1 + 1)\cdot(0.1)$

$= -1 + 0.01 = \underline{-0.99}$

$x_3 = x_2 + h = 0.3.$

$y_3 = y_2 + f(x_2, y_2)\cdot h = y_2 + (y_2 + x_2 + 1)\cdot h$

$= -0.99 + (-0.99 + 0.2 + 1)\cdot(0.1)$

$= -0.99 + 0.021$

$= \underline{-0.969}$

Problem 3. (25 points) Given the following homogeneous second-order differential equation:

$$y'' + 6y' + 13y = 0.$$

(a) Find the general solution of the given differential equation.

Solution: Chrracteristic equation.

$$r^2 + 6r + 13 = 0$$

$$r = \frac{-6 \pm \sqrt{6^2 - 4 \cdot 1 \cdot 13}}{2} = \frac{-6 \pm \sqrt{-16}}{2}$$

$$= -3 \pm 2i$$

$$\Rightarrow y(x) = e^{-3x}(C_1 \cos 2x + C_2 \sin 2x).$$

(b) Find a particular solution of the equation that satisfies the initial conditions

$$y(0) = 2, \quad y'(0) = 0.$$

Solution:
$$y'(x) = (-3)e^{-3x}(C_1 \cos 2x + C_2 \sin 2x)$$
$$+ e^{-3x}(-2C_1 \sin 2x + 2C_2 \cos 2x)$$
$$= e^{-3x}[(-3\cos 2x - 2\sin 2x)C_1 + (-3\sin 2x + 2\cos 2x)C_2]$$

$$\left.\begin{aligned} y(0) = 2 &\Rightarrow y(0) = C_1 = 2 \\ y'(0) = 0 &\Rightarrow y'(0) = -3C_1 + 2C_2 = 0 \end{aligned}\right\} \Rightarrow \begin{aligned} C_1 &= 2 \\ C_2 &= 3 \end{aligned}$$

Thus, we have

$$y(x) = e^{-3x}(2\cos 2x + 3\sin 2x)$$

Problem 4. (25 points) Given a nonhomogeneous linear third-order differential equation:

$$y^{(3)} - 4y'' + 4y' = 3x - 1.$$

(a) Find the complementary solution y_c of the above nonhomogeneous equation (i.e., the general solution of the associated homogeneous equation).

Solution: Chorcteristic equation:

$$r^3 - 4r^2 + 4r = 0$$

$$r(r^2 - 4r + 4) = 0$$

$$r(r-2)^2 = 0$$

$$\Rightarrow r_1 = 0, \quad r_2 = r_3 = 2$$

$$\Rightarrow y_c(x) = C_1 + (C_2 + C_2 x)e^{2x} = C_1 + C_2 e^{2x} + C_3 x e^{2x}$$

(b) Determine the appropriate of a particular solution y_p for the nonhomogeneous equation.

Solution: We have $f(x) = 3x - 1$, thus we set

$$y_p(x) = x^{s}(A + Bx)$$

$\hookrightarrow s = 1$ to avoid duplication of terms in y_c.

$$\Rightarrow y_p(x) = x(A + Bx)$$

$$= Ax + Bx^2$$

(c) Find all underdetermined coefficients in y_p by substituting it into the nonhomogeneous equation.

Solution: $y_p'(x) = A + 2Bx$

$$y_p''(x) = 2B$$

$$y_p^{(3)}(x) = 0$$

$$y_p^{(3)} - 4y_p'' + 4y_p' = 0 - 4(2B) + 4(A + 2Bx)$$

$$= -8B + 4A + 8Bx = 3x - 1$$

$$\Rightarrow \left.\begin{array}{l} 8B = 3 \\ -8B + 4A = -1 \end{array}\right\} \Rightarrow \begin{array}{l} A = \frac{1}{2} \\ B = \frac{3}{8} \end{array}$$

Thus, $y_p(x) = \frac{1}{2}x + \frac{3}{8}x^2$